

ENTRY 042 2,198 Circuits, the GNN Keeps Improving — August 20, 2026 — Status: COMPLETE

Entry 042 — More Data Helps the GNN Too, and a Parallelization Detour That Wasn't Worth It

1,376 to 2,198 Circuits; GNN Floor-Collapse MAE Falls Again, 6.22 to 5.31 Points

1. Growing the Dataset, and a Parallelization Attempt

Tried parallelizing circuit generation across the sandbox's 4 CPU cores via multiprocessing.Pool, keeping identical precision (4 seeds x 4096 shots, same methodology as every prior entry). It worked initially but gave only a 1.3-1.7x speedup, not 4x -- most of the per-circuit cost turned out to be Python/Qiskit overhead (circuit construction, transpile scheduling) rather than raw linear algebra, which doesn't parallelize as cleanly. It then began reliably hanging after several hundred circuits for reasons that didn't reproduce in isolated tests -- reverted to the proven single-process approach from Entries 034-040. The sandbox's own throughput also degraded sharply mid-session (from roughly 0.4 to 0.06 circuits/second) for reasons outside the model or scripts, most likely shared infrastructure load. Net result: 822 new circuits added this entry, bringing the combined dataset to 2,198 -- real progress, just slower than hoped.

2. Retraining the GNN

	1,376 circuits (Entry 041)	2,198 circuits (Entry 042)
floor-collapse cases	70	112
GNN CV R ²	0.868	0.881
GNN CV MAE (all)	2.64 pts	2.04 pts
GNN floor-collapse MAE	6.22 pts	5.31 pts
v4.1 floor-collapse MAE	17.01 pts	14.30 pts

The GNN improved on every metric with more data, same as the tabular model did in Entries 038-040 -- confirming the growth strategy pays off for the graph architecture too, not just the simpler baseline. The GNN's margin over v4.1 on floor-collapse cases remains enormous (5.31 vs 14.30 points, a 63% relative reduction, essentially unchanged from Entry 041's finding).

Entry 042 — Conclusion

The GNN is confirmed as the best model built in this project across every dataset size tested so far, and it is not yet showing diminishing returns from more data the way the closed-form model structurally can't improve at all. The path to 5,000 and eventually 10,000 circuits remains the right one; it will just take more sessions than originally hoped, since raw circuit-generation throughput -- not modeling -- is now the actual bottleneck.

3. Next Step

Continue growth toward 5,000 in future sessions using the proven sequential approach (entry042_grow_sequential.py), retraining the GNN periodically to keep tracking whether floor-collapse MAE keeps falling or starts to plateau for real this time. In parallel, build an interactive visualization of the trained GNN's message-passing process on an example circuit, showing how information flows across the real route graph toward a prediction.

Research Log — Index Addendum (Entry 042)

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Entry 042	Dataset to 2,198 Circuits; GNN Retrained and Improved	Aug 20, 2026	Complete

Scripts: entry042_grow_parallel.py (partial, superseded), entry042_grow_sequential.py (active), entry042_train_gnn.py. Data: quantumbridge_data/entry042_parallel_dataset.json, quantumbridge_data/entry042_combined_dataset.json (2,198 total), quantumbridge_data/entry042_gnn_results.json, quantumbridge_data/entry042_growth_summary.json.